



HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME

CT GROUP

CENTRE NUMBER

--	--	--	--

INDEX NUMBER

--	--	--	--

BIOLOGY

9744/04

Paper 4 Practical

28 August 2017

No additional materials are required.

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

There are **three** question booklets (**I** to **III**) to this paper. Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page and on the lines provided at the top of the cover pages of Booklets **II** and **III**.

Answer **all** questions in the spaces provided on the question paper.

Shift
Laboratory

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

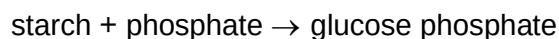
For Examiners' Use	
1	/ 25
2	/ 16
3	/ 14
Total Mark	/ 55

This document consists of **16** printed pages and **4** blank pages.

QUESTION 1

Many plants store starch in specific organs such as the bases of stems (e.g. potato tubers) or in roots before and during a dormant period. This provides the energy for growth the following year.

You are to investigate the activity of the enzyme starch phosphorylase on an extract of potato, *Solanum tuberosum*. The breakdown of stored starch within cells is catalysed by starch phosphorylase:



You are provided with 30 cm³ of potato extract, **T**. Pieces of potato tuber were chopped, mixed with a small volume of water and ground up in order to obtain **T**.

- 1** Stir **T** with a glass rod and filter it by pouring into a 100 cm³ beaker through the sieve. Place the sieve on top of the opening of the beaker to allow the filtration process to complete.
- 2** Using a small plastic spoon, obtain a spoonful of the filtration residue from the sieve and place it into another 100 cm³ beaker. Re-suspend the residue (termed as re-suspended **T**) in 30 cm³ of distilled water. Ensure that you stir thoroughly.
- 3** Using a clean plastic pipette, remove a small sample from the re-suspended **T** and place it on the white tile. Test the sample for starch and record your result below. **Using a paper towel, wipe away the sample from the white tile.**

Colour of extract on adding iodine solution: [1]

Different plant species manufacture distinctive starch grains, morphologically varying in size and shape. Each individual starch grain is composed of growth rings called lamellae and a central core called hilum.

- 4** Using the same plastic pipette from step **3**, place a few drops of re-suspended **T** on a clean glass slide. Carefully lower a coverslip over the glass slide. You are to ensure there are no air bubbles present. Now examine the mounted glass slide under the light microscope for the presence of starch grains.

Make a large labelled detailed drawing, in the space below, of two adjacent starch grains.

[4]

- 5 The potato extract was prepared by crushing and filtering some storage tissue.
Explain the result obtained in step 3 when iodine solution was added to the re-suspended T.

[1]

You are required to investigate the progress of this enzyme-catalysed reaction by **both**

- testing for the disappearance of starch, and
- testing for the appearance of glucose phosphate by finding the time taken for the decolourisation of potassium manganate(VII) solution.

To test for the production of glucose phosphate, the change in the colour of potassium manganate(VII) solution is:

purple → colourless

where the formation of a colourless solution indicates end-point.

You are provided with:

labelled	contents	hazard	volume / cm ³
E	starch phosphorylase	harmful irritant	10
S	starch solution	none	40
A	sulfuric acid	irritant	20
P	potassium manganate (VII) solution	harmful	20

You are advised to wear suitable eye protection, especially when using the starch phosphorylase solution, **E** and the sulfuric acid, **A**. If **either E** or **A** come into contact with your skin, wash off with cold water. **P** may stain your skin.

Proceed as follows:

- 6 Label the test-tubes with the sampling times of your choice. You should **not** sample for longer than 15 minutes.
- 7 Dispense 2.5 cm³ of **A** into each test-tube. Add 1 cm³ of **P** into each test-tube and gently shake to mix with **A**.

Read step 8 to step 12 and note that the stopwatch should not be stopped until you have the last end-point recorded. *The reaction will start as soon as E is added to S.*

- 8 Dispense 30 cm³ of **S** into a plastic vial. Add 4 cm³ of **E** into the plastic vial containing **S**. Immediately stir the mixture in the beaker and start the timer.
- 9 At each of your sampling times,
- test for the disappearance of starch as described in step 3, and
 - test for the appearance of glucose phosphate by removing 3 cm³ of the mixture and putting it into the appropriately labelled test-tube, mixing well.

- 10** Record, in **(a)** below, the time shown on the stopwatch when the colourless end-point is reached (**raw** result). Do **not** stop the stopwatch.
- 11** Repeat step **9** for each of the times decided in step **6** until you have recorded the end-point for the last sample, removed at 15 minutes.
- 12** Calculate the time taken to reach **each** end-point (**processed** results).
- (a)** Prepare the space below to record your results for
- the test for starch,
 - the **raw** results for the appearance of glucose phosphate, and
 - your **processed** results for the appearance of glucose phosphate.

[5]

(b) Account for the results obtained in **(a)**.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[3]

- (c) Identify **one** limitation of this investigation and suggest a way in which the experiment can be improved to give more accurate and reliable results.

.....

.....

.....

..... [2]

- (d) This procedure investigated the progress of the hydrolysis of starch by starch phosphorylase. To modify this procedure for investigating a different variable, the time for the hydrolysis would be standardised.

Consider how you could modify this procedure to investigate the effect of **pH** on the activity of starch phosphorylase.

Describe how the independent variable, **pH**, could be investigated.

.....

.....

.....

.....

.....

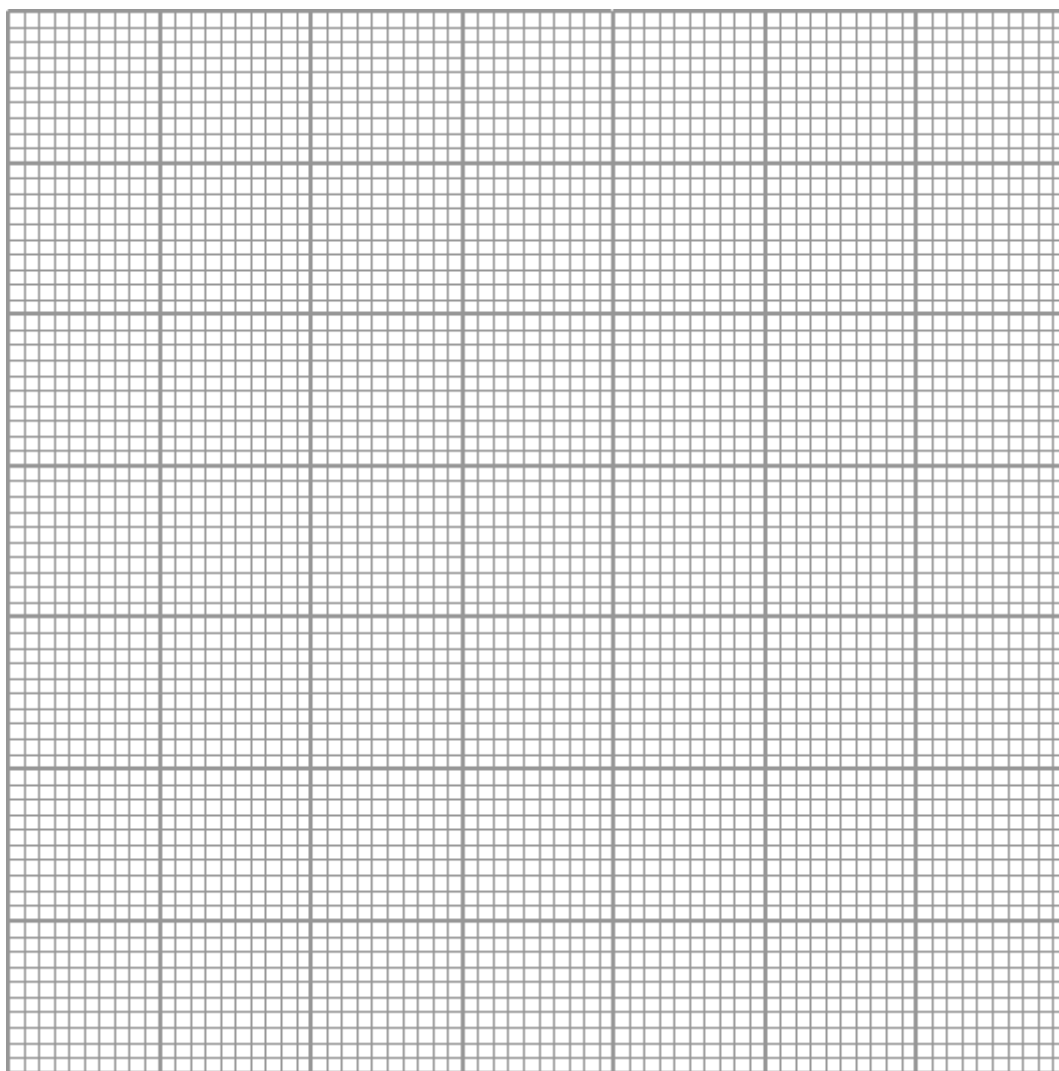
..... [3]

- (e) A student investigated the effect of starch concentration on the initial rate of reaction of starch phosphorylase. Table 1.1 shows the results for this investigation.

Table 1.1

percentage concentration of starch	initial rate of reaction of starch phosphorylase / arbitrary units
0.00	0
0.05	100
1.25	215
1.75	285
2.50	340
3.25	340

- (i) Use the grid provided to plot a graph of the data shown in Table 1.1.



[4]

- (ii) Use the graph to estimate the Michaelis-Menten constant (K_m).
Show your working on the graph and in the space below.

$K_m =$ [2]

[Total: 25]

BLANK PAGE

Candidate Name: _____

CT Group: 16S7 _____

BOOKLET II**BLANK PAGE**

QUESTION 2

During this question you will require access to a microscope, slide **S1** and a stage micrometer.

Leaves require carbon dioxide and water for photosynthesis. For carbon dioxide to enter, the stomata on the surface of leaves must be open. However, the plant must not lose too much water. The plant must strike a balance between conserving water and bringing in sufficient amounts of carbon dioxide for photosynthesis.

S1 is a slide of a stained transverse section through a xerophytic plant leaf, which is adapted to conserve water.

You are not expected to be familiar with this specimen.

- (a)(i)** Examine the slide under the low-power objective lens of your microscope. Observe the cells found immediately below the upper epidermis in the shaded area shown in Fig. 2.1.



Fig. 2.1

Select one group of **four** cells found immediately below the upper epidermis. Examine this group of cells under the high-power objective lens. Each cell in the group should touch two of the other cells and at least one cell should be capable of making starch.

Make a large labelled detailed drawing, in the space below, of this group of **four** cells.

[5]

- (ii) Examine slide **S1** under the low-power objective lens of your microscope again. Make a large plan drawing of the part of the leaf indicated by the shaded area in Fig. 2.1.

On your drawing, use **one** ruled label line and label to identify one feature that adapts the plant to living in a dry habitat.

Annotate this label to explain **how** the feature you have identified adapts this plant to living in a dry habitat.

[3]

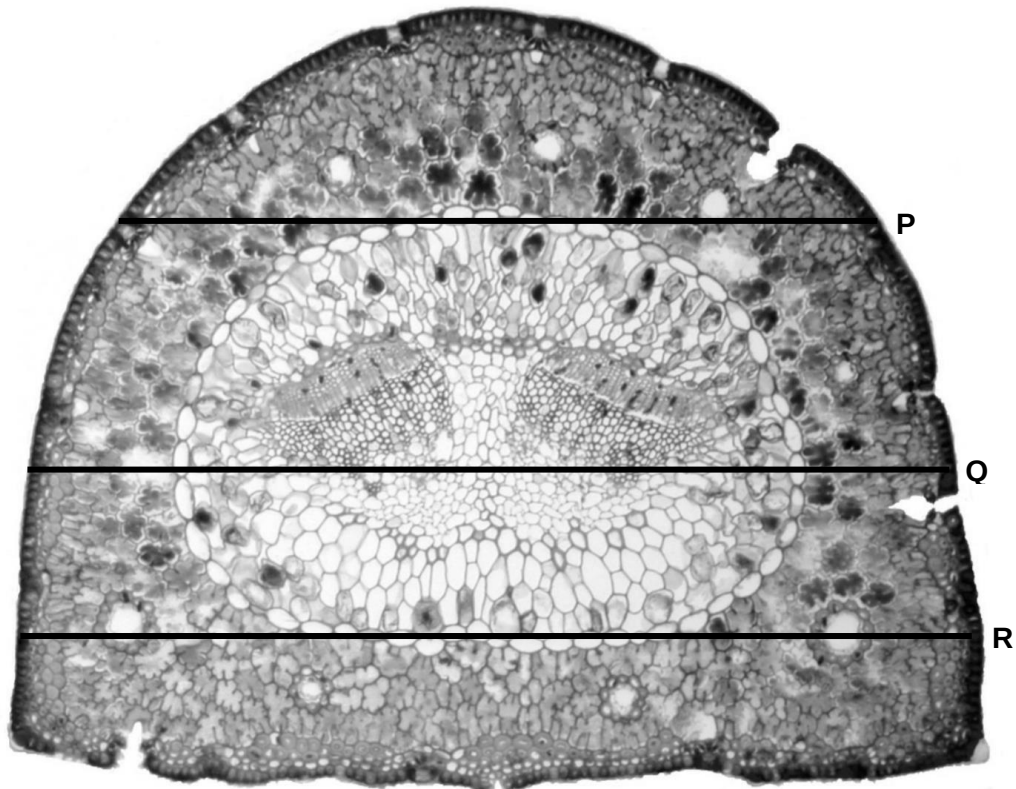
- (iii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual width, in μm , of the leaf.

Show the measurements that you made and your working.

width of leaf = μm [2]

- (b) Fig. 2.2 is a photomicrograph of a stained transverse section through a leaf of a different xerophytic plant species.

You are not expected to be familiar with this specimen.



Magnification $\times 175$

Fig. 2.2

Use the magnification and the lines in Fig. 2.2 to find the actual width of the leaf, in μm , at positions labelled **P**, **Q**, and **R**.

Show the measurement that you made and your working for the width at position **Q**.

P μm , **Q** μm , **R** μm [2]

- (c) Leaves of xerophytic plants have morphological adaptations that facilitate conservation of water.

A student hypothesises that xerophytic plants have narrower leaves than mesophytic plants, which grow in areas where water is more readily available. The student measured the width of 30 leaves from a mesophytic plant species, **X**, and that of 30 leaves from a xerophytic plant species, **Y**.

- (i) State how narrower leaves allow xerophytic plants to conserve water.

.....
 [1]

A summary of the student's results is shown in Table 2.1.

Table 2.1

plant species	width of leaves / μm		
	mean, $\bar{x}$	standard deviation, s	$\bar{x} \pm 2s$
X	1012.5	186.1	1012.5 ± 372.2
Y	562.0	14.8	562.0 ± 29.6

- (ii) Comment on what these results show and explain if the results support the student's hypothesis.

.....

 [3]

[Total: 16]

Candidate Name: _____

CT Group: 16S7 _____

BOOKLET III

BLANK PAGE

QUESTION 3

The rate of respiration of an organism is an indication of its demand for energy. Respiration rate may be measured by means of a respirometer.

A student used a respirometer to compare the rate of respiration amongst three organisms

- single-celled green algae, immobilised in sodium alginate beads,
- germinating seeds, and
- insect larvae.

After putting the single-celled green algae, immobilised in sodium alginate beads, into the air-filled container and attaching the graduated tube, the respirometer was lowered into a water trough as shown in Fig. 3.1. The green algae respired and water moved into the graduated tube. The procedure was repeated for the other two organisms.

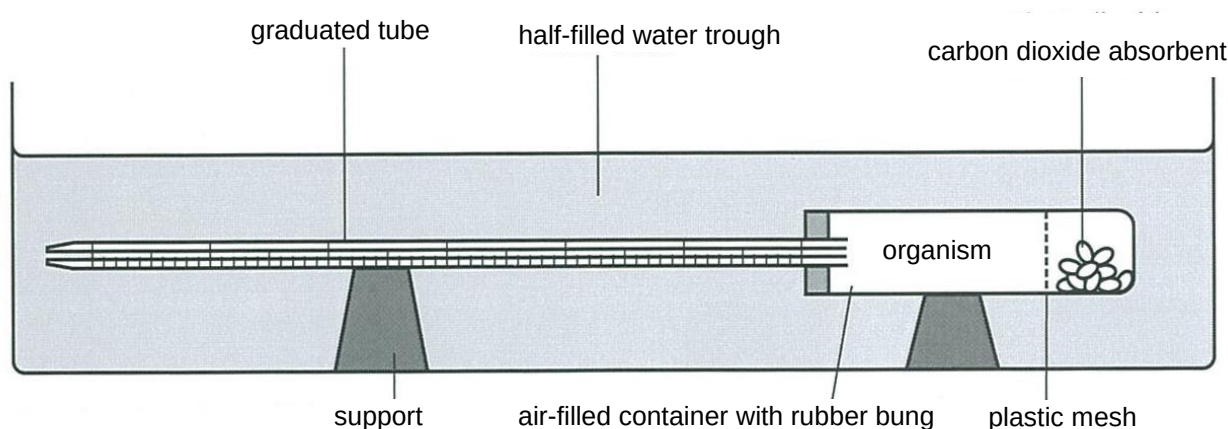


Fig. 3.1

Using this information and your own knowledge, design an experiment to compare the rates of respiration of single-celled green algae, germinating seeds and insect larvae.

You must use:

- single-celled green algae, immobilised in sodium alginate
- germinating seeds
- insect larvae
- half-filled water trough
- carbon dioxide absorbent
- air-filled container with rubber bung
- graduated tube
- weighing balance

You may select from the following apparatus:

- normal laboratory glassware, e.g. boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- thermometer
- petroleum jelly
- stopwatch
- marker
- white card

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Copyright Acknowledgments:

© Hwa Chong Institution 2017